

Section I

5 marks

Attempt Questions 1-5

Allow about 5 minutes for this section

Use the multiple choice answer sheet for questions 1-5

1. Which of the following points lies on the line described by the vector equation

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$$

(A) $\begin{pmatrix} -3 \\ 9 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} -3 \\ -8 \\ -3 \end{pmatrix}$

(C) $\begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$

(D) $\begin{pmatrix} 3 \\ -5 \\ 2 \end{pmatrix}$

2. If the vectors $\vec{u} = \lambda \vec{i} + \lambda \vec{j} + 2\vec{k}$ and $\vec{v} = \lambda \vec{i} - 2\vec{j} - 4\vec{k}$ are perpendicular, then:

(A) $\lambda = -2$ or $\lambda = 4$

(B) $\lambda = -2$ or $\lambda = -4$

(C) $\lambda = 2$ or $\lambda = -4$

(D) $\lambda = 2$ or $\lambda = 4$

3. P , Q and R are points with position vectors $\vec{p}$, $\vec{q}$ and $\vec{r}$ respectively.

The points are collinear and P lies between R and Q . If $|QR| = 3|PR|$, then the position vector of $\vec{r}$ is:

(A) $\vec{r} = \frac{1}{2}\vec{p} - \frac{3}{2}\vec{q}$

(B) $\vec{r} = \frac{3}{2}\vec{p} - \frac{1}{2}\vec{q}$

(C) $\vec{r} = \frac{3}{2}\vec{p} + \frac{1}{2}\vec{q}$

(D) $\vec{r} = \frac{3}{2}\vec{q} - \frac{1}{2}\vec{p}$

4. A particle is moving in SHM along a straight line. At time t seconds it has displacement x metres from a fixed point on the line and velocity v metres/second given by $v^2 = -0.5x^2 + 2x + 2.5$.

What is the period of the motion?

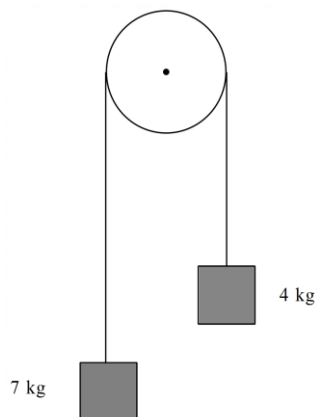
(A) π seconds

(B) $\pi\sqrt{2}$ seconds

(C) 2π seconds

(D) $2\pi\sqrt{2}$ seconds

5. A light, inextensible string passes over a smooth pulley. Attached to the ends of the string are masses of 4 kg and 7 kg, as shown.



The acceleration in the downwards direction of the 7 kg mass is:

- (A) $\frac{3g}{11}$
(B) $\frac{11g}{3}$
(C) $\frac{7g}{11}$
(D) $3g$

End of Section I

Section II

36 marks

Attempt Questions 6-8

Allow about 55 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (Start a new writing booklet.)

Marks

- a. Find the centre and radius of the sphere with equation: 3

$$x^2 + y^2 + z^2 - x + 2y + 3z = 1$$

- b. Relative to the origin, O, the points A , B , C and D have position vectors given by: 3

$$A: -4\underset{\sim}{i} + 3\underset{\sim}{j} + 3\underset{\sim}{k}$$

$$B: 4\underset{\sim}{i} - 2\underset{\sim}{j} + 6\underset{\sim}{k}$$

$$C: 4\underset{\sim}{i} - \underset{\sim}{j} - \underset{\sim}{k}$$

$$D: 2\underset{\sim}{j} - 6\underset{\sim}{k}$$

Find the position vector of the point of intersection of the lines AC and BD .

- c. Given that the vectors $\underset{\sim}{u}$ and $\underset{\sim}{v}$ satisfy $\underset{\sim}{u} = 9\underset{\sim}{i} + 4\underset{\sim}{j} - \underset{\sim}{k}$ and $\underset{\sim}{v} = 8\underset{\sim}{i} - 5\underset{\sim}{j} + 3\underset{\sim}{k}$, find the acute angle between the vectors $\underset{\sim}{u}$ and $\underset{\sim}{v}$. 3

Question 6 continued on next page.

Question 6 continued.

d. Consider the points:

$$A = (4, 3, -2), B = (-3, -6, 10) \text{ and } C = (-38, -51, -49)$$

- i) Find the equation of the line, r , which passes through A and B . 1
- ii) Show that A , B and C are not collinear. 2

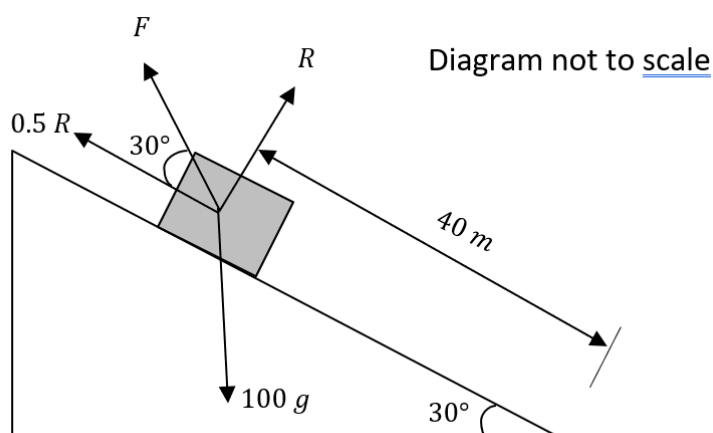
End of Question 6

Question 7 (Start a new writing booklet.)**Marks**

- a. An object with mass 100 kg begins at rest and slides down a rough surface which is inclined at 30° to the horizontal.

The object experiences a resistive force of $0.5R$ Newtons, where R is the normal reaction force exerted by the surface on the object. In order to slow the object down, a force F is applied to the object at an angle of 30° to the plane, as shown in the diagram.

Take the acceleration due to gravity, g , to be 10 m/s^2 .



- i) By resolving forces perpendicular to the plane, show that 2

$$R = 500\sqrt{3} - \frac{F}{2}$$

- ii) Show that the acceleration of the object **down** the plane is given by: 2

$$a = \frac{1000(2-\sqrt{3})-F(2\sqrt{3}-1)}{400}$$

Question 7 continued on next page.

Question 7 continued.

- b. A particle is projected from the Origin with an initial speed of V at an angle of θ above the horizontal. The particle is affected by both gravity and an air resistance proportional to the velocity such that the components of acceleration are:

$$\begin{aligned}\ddot{x} &= -k\dot{x} \\ \ddot{y} &= -k\dot{y} - g\end{aligned}$$

Where k is a constant and $g \text{ m/s}^2$ is the acceleration due to gravity.

- i) Show that: 3

$$\begin{aligned}\dot{x} &= V \cos \theta e^{-kt} \quad \text{and,} \\ \dot{y} &= \frac{1}{k}(g + kV \sin \theta)e^{-kt} - \frac{g}{k}\end{aligned}$$

- ii) Show that the particle reaches maximum height when 2

$$t = \frac{1}{k} \ln \left(\frac{g + kV \sin \theta}{g} \right)$$

- c. A mass of 1 kg moves along a straight line with velocity $v \text{ m/s}$. The object has initial velocity $U \text{ m/s}$, where $U > 0$, and starts at the Origin. It encounters a resistance of $v + v^3 \text{ m/s}^2$ such that $\ddot{x} = -v(1 + v^2)$.

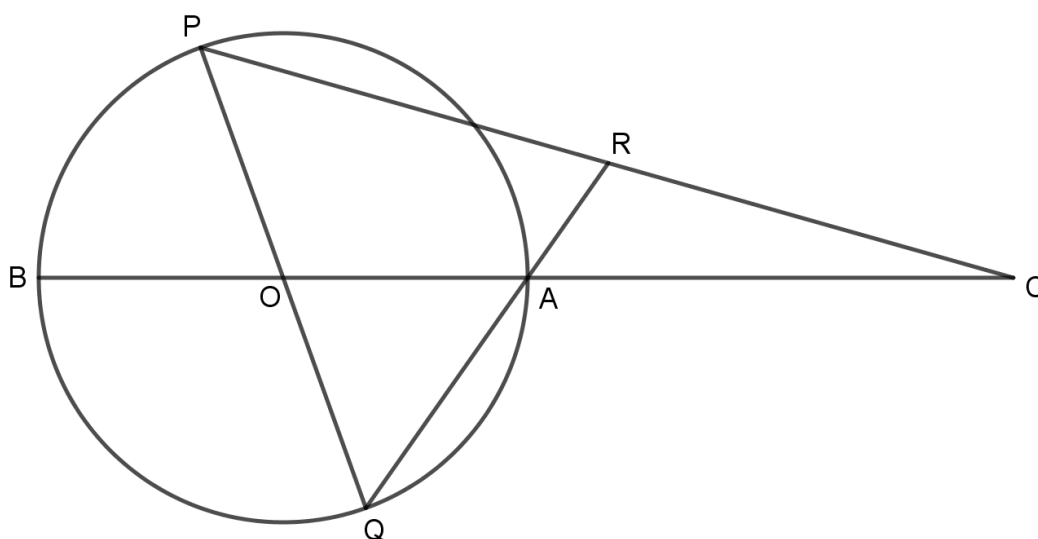
Show that $x = \tan^{-1} \left(\frac{U-v}{1+Uv} \right)$ 3

End of Question 7

Question 8 (Start a new writing booklet.)**Marks**

- a. Calculate the length of the vector $2\vec{a} - 3\vec{b}$, given that $|\vec{a}| = 3$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 4$ 2

b.



In the diagram, PQ and AB are diameters of a circle with centre O . BA is produced to C so that $BA = AC$. QA produced meets PC in R . Let $\vec{OA} = \vec{a}$ and $\vec{OP} = \vec{p}$.
Given $\vec{PR} = \lambda \vec{PC}$ and $\vec{QR} = \mu \vec{QA}$ for some scalars λ and μ .

- i) Find expressions, in terms of $\vec{a}$ and $\vec{p}$, for $\vec{PR}$ and $\vec{QR}$. 2
- ii) Show that R is the midpoint of PC . 2

Question 8 continued on next page.

Question 8 continued.

- c. A particle of mass m kg falls vertically from rest under gravity in a medium where the resistance to motion has magnitude $\frac{mv^2}{g}$ Newtons when the speed of the particle is v m/s and the acceleration due to gravity is g m/s². At time t seconds the particle has fallen x metres and has velocity v m/s.

i) Show that $\ddot{x} = \frac{1}{g}(g^2 - v^2)$ 1

ii) Show that $v = g \left(\frac{e^{2t}-1}{e^{2t}+1} \right)$. 3

You may wish to use the result $\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left(\frac{a+x}{a-x} \right) + c$

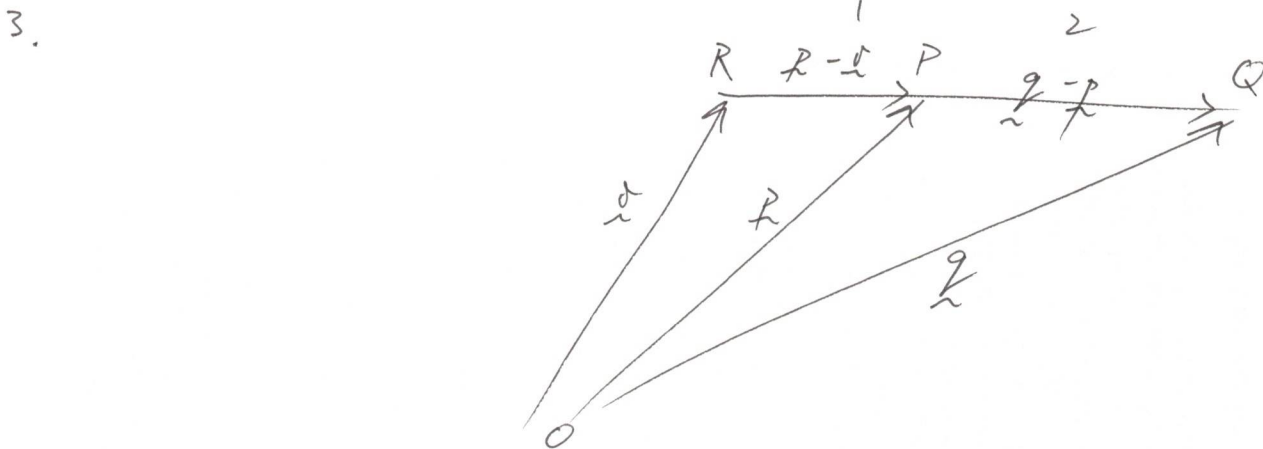
- iii) Find in simplest exact form, the time taken for the particle to reach half of its terminal velocity. 2

End of Question 8

End of Paper

$$1. \quad \lambda = -1 \Rightarrow \underline{U} = \begin{pmatrix} 1+2 \\ 2-3 \\ -1-1 \end{pmatrix} \\ = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} \Rightarrow \textcircled{C}$$

$$2. \quad \underline{U} \cdot \underline{V} = 0 \\ \text{ie } \lambda^2 - 2\lambda - 8 = 0 \\ (\lambda + 2)(\lambda - 4) = 0 \\ \lambda = -2 \text{ or } \lambda = 4 \quad \textcircled{A}$$



$$\underline{q} - \underline{r} = 2(\underline{p} - \underline{r}) \\ \underline{q} - \underline{r} = 2\underline{p} - 2\underline{r} \\ 2\underline{r} = 3\underline{p} - \underline{q} \\ \underline{r} = \frac{3}{2}\underline{p} - \frac{1}{2}\underline{q} \quad \textcircled{D}$$

4.

$$v^2 = -\frac{x^2}{2} + 2x + \frac{5}{2}$$

$$= \frac{1}{2} (5 - (x^2 - 4x))$$

$$= \frac{1}{2} (5 - (x^2 - 4x + 4) + 4)$$

$$= \frac{1}{2} (9 - (x-2)^2)$$

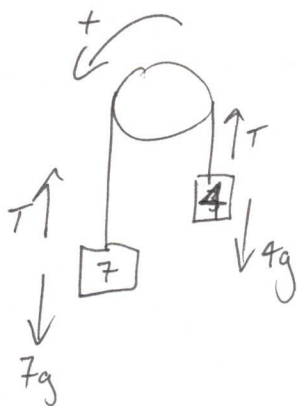
$$v^2 = n^2 (a^2 - x^2) \quad \therefore n = \frac{1}{\sqrt{2}}$$

$$T = \frac{2\pi}{n}$$

$$= 2\sqrt{2} \pi$$

(D)

5.



$$\sum F = M\ddot{x} = 7g - T + T - 4g$$

$$11\ddot{x} = 3g$$

$$\ddot{x} = \frac{3g}{11}$$

(A)

6. a.

$$x^2 + y^2 + z^2 - x + 2y + 3z = 1$$

Some attempt. ①

$$x^2 - x + \left(\frac{1}{2}\right)^2 + y^2 + 2y + 1 + z^2 + 3z + \left(\frac{3}{2}\right)^2 = 1 + \frac{1}{4} + 1 + \frac{9}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + (y + 1)^2 + \left(z + \frac{3}{2}\right)^2 = \frac{9}{2}$$

$$\therefore \text{Centre is } \left(\frac{1}{2}, -1, -\frac{3}{2}\right) \quad \text{①}$$

$$\text{Radius is } \frac{3\sqrt{2}}{2} \text{ units} \quad \text{①}$$

$$b. \quad \vec{r}_{AC} = \begin{pmatrix} -4 \\ 3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix}$$

① Some progress

$$\vec{r}_{BD} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 4 \\ -12 \end{pmatrix}$$

At pt. of int.

$$-4 + 8\lambda = 4 - 4\mu \quad (1)$$

$$3 - 4\lambda = -2 + 4\mu \quad (2)$$

$$3 - 4\lambda = 6 - 12\mu \quad (3)$$

① Evidence

$$\text{From (2), (3)} \quad -2 + 4\mu = 6 - 12\mu$$

$$16\mu = 8$$

$$\mu = \frac{1}{2}$$

$$\therefore \text{Pt of int is } \begin{pmatrix} 4 - \frac{1}{2} \times 4 \\ -2 + \frac{1}{2} \times 4 \\ 6 - \frac{1}{2} \times 12 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \quad (1)$$

$$c. \quad \vec{u} + \vec{v} = 17\vec{i} - \vec{j} + 2\vec{k} \quad (1)$$

$$\vec{u} - \vec{v} = \vec{i} + 9\vec{j} - 4\vec{k} \quad (2)$$

$$(1) + (2) \Rightarrow 2\vec{u} = 18\vec{i} + 8\vec{j} - 2\vec{k}$$

$$\vec{u} = 9\vec{i} + 4\vec{j} - \vec{k}$$

$$(1) - (2) \Rightarrow 2\vec{v} = 16\vec{i} - 10\vec{j} + 6\vec{k}$$

$$\vec{v} = 8\vec{i} - 5\vec{j} + 3\vec{k}$$

$$|\underline{u}| = \sqrt{9^2 + 4^2 + 1^2}$$

$$= \sqrt{98}$$

①

$$|\underline{v}| = \sqrt{8^2 + 5^2 + 3^2}$$

$$= \sqrt{98}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$$\textcircled{1} \quad 9 \cdot 8 - 4 \cdot 5 - 1 \cdot 3 = 98 \cos \theta$$

$$49 = 98 \cos \theta$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3} \quad \textcircled{1}$$

$$d. \quad 1) \quad \underline{r} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -7 \\ -9 \\ 12 \end{pmatrix} \quad \textcircled{1}$$

ii) If C lies on $\underline{r}$ then

$$\begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -7 \\ -9 \\ 12 \end{pmatrix} = \begin{pmatrix} -38 \\ -51 \\ -49 \end{pmatrix} \quad \textcircled{1}$$

$$\text{ie } 4 - 7\lambda = -38 \quad 3 - 9\lambda = -51 \quad -2 + 12\lambda = -49$$

$$42 = 7\lambda$$

$$54 = 9\lambda$$

$$12\lambda = -47$$

$$\lambda = 6$$

$$\lambda = 6$$

$$\lambda = -\frac{47}{12}$$

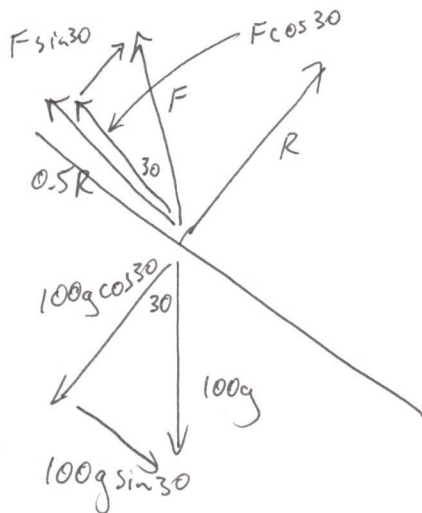
①

$$\neq 6$$

Since the λ value is not the same for all components, C does not lie on $\overrightarrow{AB}$.

7a.

i)



$\perp$ to plane $R + F \sin 30 = 100g \cos 30$ ①

$$R = 100 \times 10 \times \frac{\sqrt{3}}{2} - F \times \frac{1}{2}$$
 ①

$$= 500\sqrt{3} - \frac{F}{2}$$

ii) $\sum F = m\ddot{x} = 100g \sin 30 - \frac{R}{2} - F \cos 30$ ①

$$100a = 1000 \times \frac{1}{2} - \left(\frac{500\sqrt{3}}{2} - \frac{F}{4} \right) - F \times \frac{\sqrt{3}}{2}$$

$$= 500 - \frac{500\sqrt{3}}{2} + \frac{F}{4} - \frac{F\sqrt{3}}{2}$$

$$= \frac{2000 - 1000\sqrt{3} + F - 2F\sqrt{3}}{4}$$
 ①

$$a = \frac{1000(2 - \sqrt{3}) + F(2\sqrt{3} - 1)}{400}$$

b.



$$1) \quad \ddot{x} = -kx$$

$$\frac{dx}{dt} = -kx$$

$$\frac{dt}{dx} = -\frac{1}{k} \times \frac{1}{x}$$

$$\int_0^t dt = -\frac{1}{k} \int_{v \cos \theta}^x \frac{1}{x} dx$$

$$t = -\frac{1}{k} \left[\ln x \right]_{v \cos \theta}^x$$

$$-kt = \ln \left(\frac{x}{v \cos \theta} \right)$$

$$e^{-kt} = \frac{x}{v \cos \theta}$$

①

$$x = v \cos \theta e^{-kt}$$

$$\ddot{y} = -ky - g$$

$$\frac{dy}{dt} = -ky - g$$

$$\frac{dt}{dy} = \frac{-1}{ky + g}$$

$$\int_0^t dt = -\frac{1}{k} \int_{v \sin \theta}^y \frac{k}{ky + g} dy$$

①

$$t = -\frac{1}{k} \left[\ln(k\dot{y} + g) \right]_{V\sin\theta}^{\dot{y}}$$

$$-kt = \ln\left(\frac{k\dot{y} + g}{kV\sin\theta + g}\right)$$

$$e^{-kt} = \frac{k\dot{y} + g}{kV\sin\theta + g}$$

$$(g + kV\sin\theta) e^{-kt} = k\dot{y} + g$$

(1)

$$\dot{y} = \frac{1}{k} (g + kV\sin\theta) e^{-kt} - \frac{g}{k}$$

ii) At max height, $\dot{y} = 0$

$$\therefore \frac{g}{k} = \frac{1}{k} (g + kV\sin\theta) e^{-kt}$$

(1)

$$\frac{g}{g + kV\sin\theta} = e^{-kt}$$

$$\ln\left(\frac{g}{g + kV\sin\theta}\right) = -kt$$

$$t = -\frac{1}{k} \ln\left(\frac{g}{g + kV\sin\theta}\right)$$

$$= \frac{1}{k} \ln\left(\frac{g + kV\sin\theta}{g}\right)^{-1}$$

$$= \frac{1}{k} \ln\left(\frac{g}{g + kV\sin\theta}\right)$$

(1)

c.

$$\ddot{x} = -v(1+v^2)$$

$$v \frac{dv}{dx} = -v(1+v^2)$$

$$\frac{dx}{dv} = \frac{-1}{1+v^2}$$

$$\int_0^x dx = - \int_U^v \frac{1}{1+v^2} dv \quad (1)$$

$$x = - \left[\tan^{-1} v \right]_U^v$$

$$= -(\tan^{-1} v - \tan^{-1} U)$$

$$= \tan^{-1} U - \tan^{-1} v \quad (1)$$

$$\text{let } \tan^{-1} U = \alpha \quad \tan^{-1} v = \beta$$

$$U = \tan \alpha \quad v = \tan \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$= \frac{U - v}{1 + Uv}$$

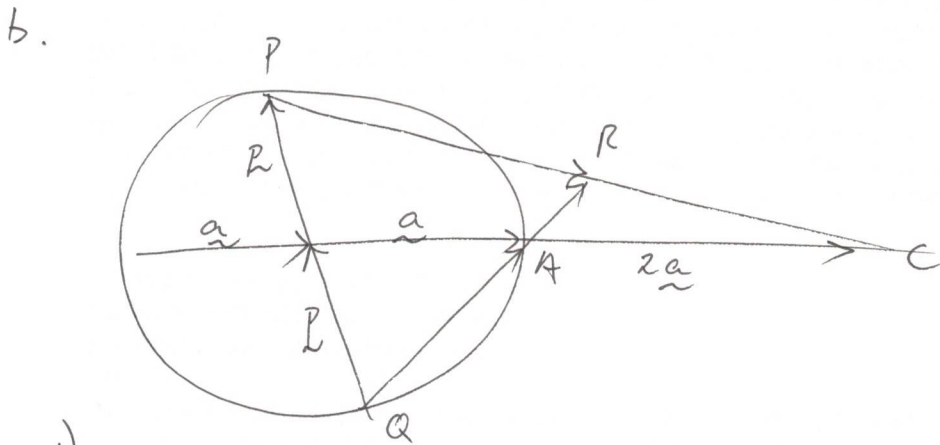
$$\therefore \alpha - \beta = \tan^{-1} \left(\frac{U - v}{1 + Uv} \right)$$

$$\text{ie } \tan^{-1} U - \tan^{-1} v = \tan^{-1} \left(\frac{U - v}{1 + Uv} \right)$$

$$\text{ie } x = \tan^{-1} \left(\frac{U - v}{1 + Uv} \right) \quad (1)$$

$$\begin{aligned} 8a. \quad |2\underline{a} - 3\underline{b}|^2 &= (2\underline{a} - 3\underline{b}) \cdot (2\underline{a} - 3\underline{b}) \\ &= 4\underline{a} \cdot \underline{a} - 12\underline{a} \cdot \underline{b} + 9\underline{b} \cdot \underline{b} \quad (1) \\ &= 4|\underline{a}|^2 - 12 \times 4 + 9|\underline{b}|^2 \\ &= 4 \times 9 - 48 + 9 \times 4 \\ &= 24 \end{aligned}$$

$$\therefore |2a - 3b| = \sqrt{24} = 2\sqrt{6} \quad (1)$$



$$\begin{aligned} \vec{PC} &= 3\hat{a} - \hat{p} \\ \vec{PR} &= \lambda(3\hat{a} - \hat{p}) \end{aligned} \quad (1)$$

$$\vec{QA} = \underline{a} + \underline{r}$$

$$\vec{QR} = \mu(\underline{a} + \underline{p}) \quad (1)$$

ii) Now, $2\vec{P} + \vec{PR} = \vec{QR}$

$$2\vec{P} + \lambda(3\vec{a} - \vec{P}) = \mu(\vec{a} + \vec{P}) \quad (1)$$
$$2\vec{P} - \lambda\vec{P} + 3\lambda\vec{a} = \mu\vec{a} + \mu\vec{P}$$

$$\therefore 3\lambda = \mu \quad \text{and} \quad 2-\lambda = \mu$$

$$\therefore 2-\lambda = 3\lambda$$

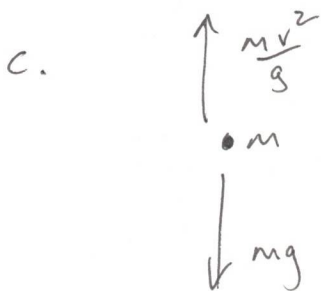
$$2 = 4\lambda$$

$$\lambda = \frac{1}{2}$$

①

$$\therefore \vec{PR} = \frac{1}{2} \times \vec{PC}$$

ie R is the midpoint of PC



$$i) \quad m\ddot{r} = mg - \frac{mv^2}{g}$$

$$\ddot{r} = g - \frac{v^2}{g}$$

$$= \frac{1}{g}(g^2 - v^2)$$

①

$$ii) \quad \frac{dv}{dt} = \frac{1}{g}(g^2 - v^2)$$

$$\frac{dt}{dv} = \frac{g}{g^2 - v^2}$$

$$\int_0^* dt = g \int_0^v \frac{1}{g^2 - v^2} dv$$

①

$$* = g \left[\frac{1}{2g} \ln \left(\frac{g+v}{g-v} \right) \right]_0^v$$

$$x = \frac{1}{2} \left[\ln \left(\frac{g+v}{g-v} \right) - \ln \left(\frac{g}{g} \right) \right]$$

$$= \frac{1}{2} \ln \left(\frac{g+v}{g-v} \right)$$

$$e^{2x} = \frac{g+v}{g-v} \quad (1)$$

$$(g-v)e^{2x} = g+v$$

$$ge^{2x} - ve^{2x} = g+v$$

$$ge^{2x} - g = ve^{2x} + v$$

$$g(e^{2x} - 1) = v(e^{2x} + 1) \quad (1)$$

$$v = \frac{g(e^{2x} - 1)}{(e^{2x} + 1)}$$

iii) Terminal vel, v_T when $\ddot{x} = 0$

$$\text{i.e. } g = \frac{v_T^2}{g}$$

$$v_T = g \quad (1)$$

$$\text{Halt } v_T = \frac{g}{2}$$

$$\therefore x = \frac{1}{2} \ln \left(\frac{g + \frac{g}{2}}{g - \frac{g}{2}} \right)$$

$$= \frac{1}{2} \ln \left(\frac{\frac{3g}{2}}{\frac{g}{2}} \right)$$

$$= \frac{1}{2} \ln \left(\frac{3g}{2} \times \frac{2}{g} \right)$$

$$= \frac{1}{2} \ln 3$$

①

ie time to $\frac{V_T}{2}$ is $\frac{1}{2} \ln 3$ sec.